

IBM HR Analytics: Understanding and Reducing Employee Attrition

Introduction

This report presents an analysis of employee attrition at IBM using HR data containing 1,470 employee records. The objective is to identify key factors driving employee resignation and provide actionable recommendations to reduce turnover.

Data Overview

Fields	Details
Source	IBM HR Analytics Dataset (Kaggle)
Total Records	1,470
Total Features	32
Target Variable	Attrition (Yes/No)
Attrition Rate	16.1%

Data Cleaning Process

Columns Removed

EmployeeCount, Over18, StandardHours These features do not provide any variability and therefore do not contribute to explaining or predicting employee attrition.

Data Type Conversion

Attrition, OverTime changed to 0/1.
Categorical columns converted to category type.

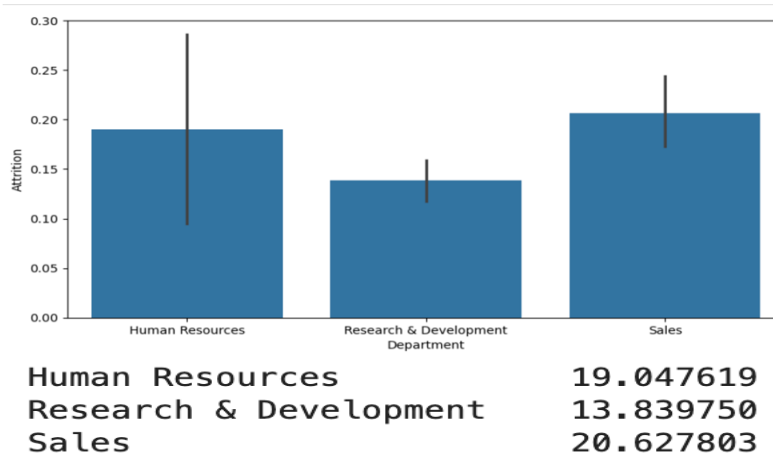
Gived Alias/Label

Performance Rating, Job Satisfaction etc → Low/Medium/High/Very High

EDA

Which department has the highest attrition rate, and how significant is the difference?

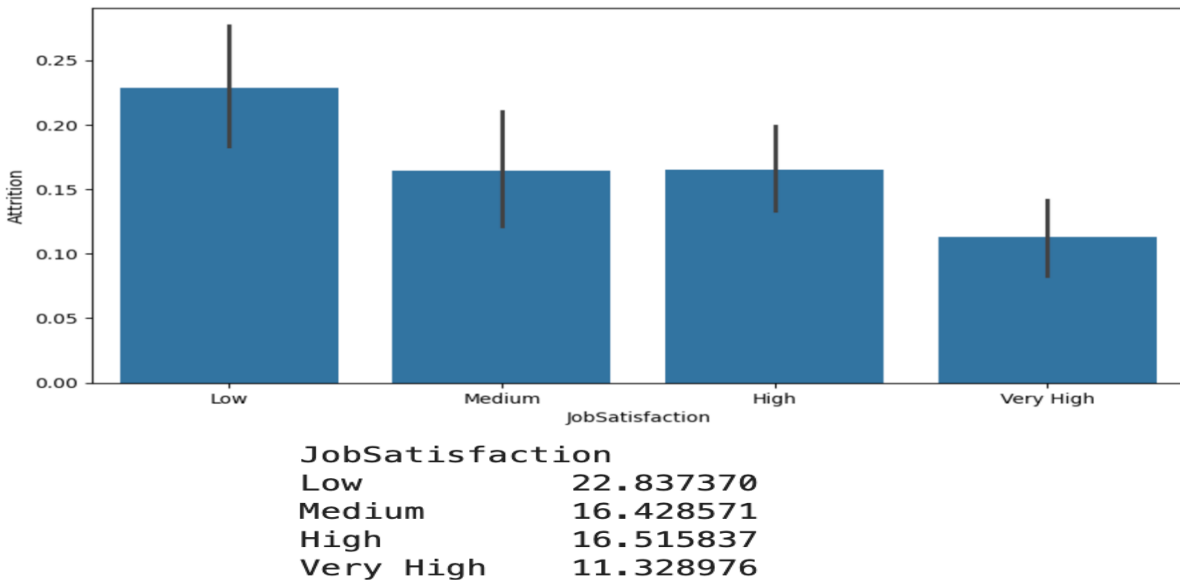
The bar chart shows that the Attrition rate of three department. Where Sales department have highest Attrition rate 20.62% which is highest and the lowest attrition rate are in the Research and development department.



Sales employees face high target pressure, commission-based salaries that create income instability, and stressful customer-facing role all of which contribute to a higher likelihood of resignation compared to other department.

Does job satisfaction level impact employee attrition rate?

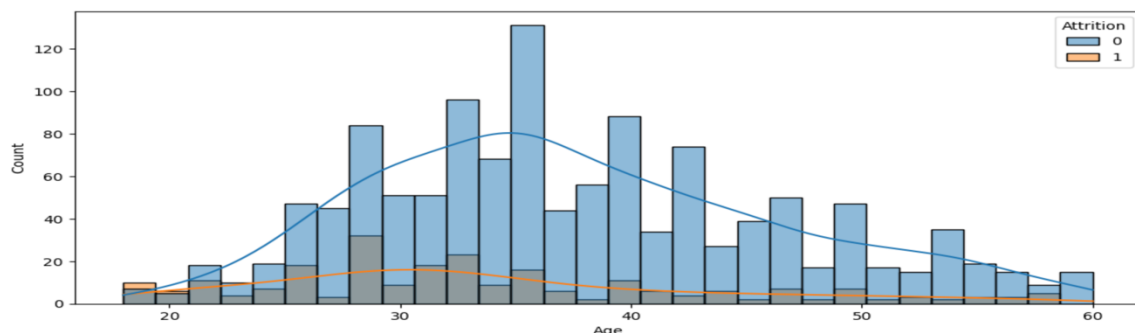
The bar chart shows that the attrition rate of Job Satisfaction. Where the attrition rate of low Satisfaction(22.83%) is double than the very high job satisfaction(11.32%).



Employees with low job satisfaction are more likely to feel disengaged and unmotivated in their role. This dissatisfaction drive them to seek better opportunities elsewhere, resulting in higher resignation. In contrast, employees who find their work fulfilling tend to remain committed to the organization.

Do younger employees resign more than older ones?

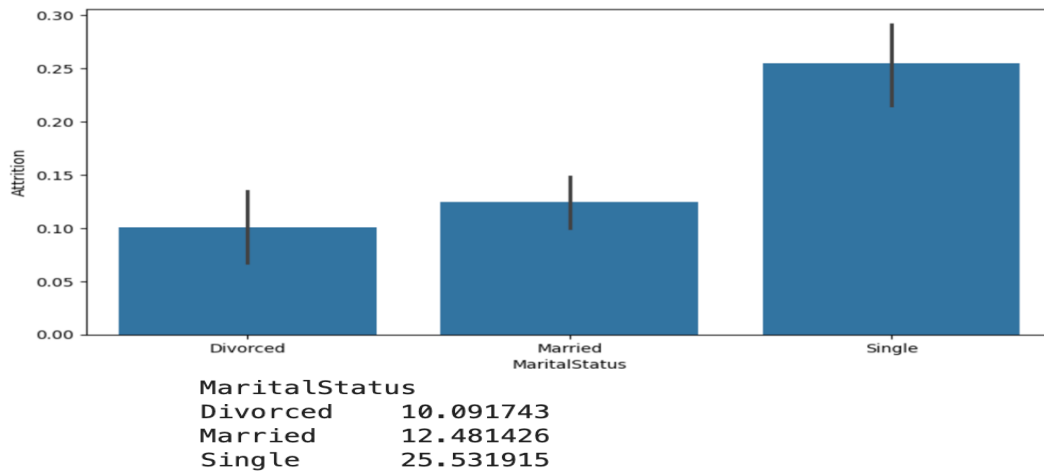
The histplot shows that younger employees aged 18-25 have the highest rate 35% followed by 25-35 age group at 19%. As age increase, the attrition rate decreases significantly dropping to 9% for employees aged 35-45.(% from Tableau)



Younger employees are more willing to take career risk as they have fewer personal responsibilities such as family obligations and financial commitments. This freedom encourages them to actively seek better opportunities, making them more likely to resign, in pursuit of career growth.

Do unmarried employees have a higher attrition rate than married employees?

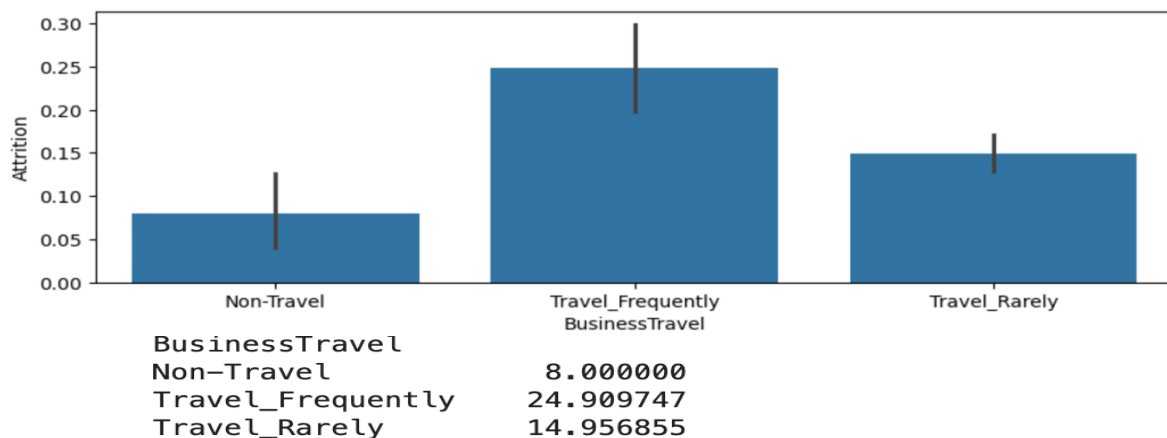
This bar chart shows that the employees whose marital status is single have the highest attrition rate(25.53%) followed by married employees (12.48%) while the divorced employees have the lowest attrition rate(10%).



Single employees have fewer personal and financial responsibilities, which gives them the confidence to take career risks in pursuit of a better life. Married employees, on the other hand, prioritize stability due to family obligations, resulting in a lower attrition rate. Divorced employees face significant financial commitments, making resignation a risky decision — which explains why they have the lowest attrition rate among all groups.

Business Travel effect on Attrition?

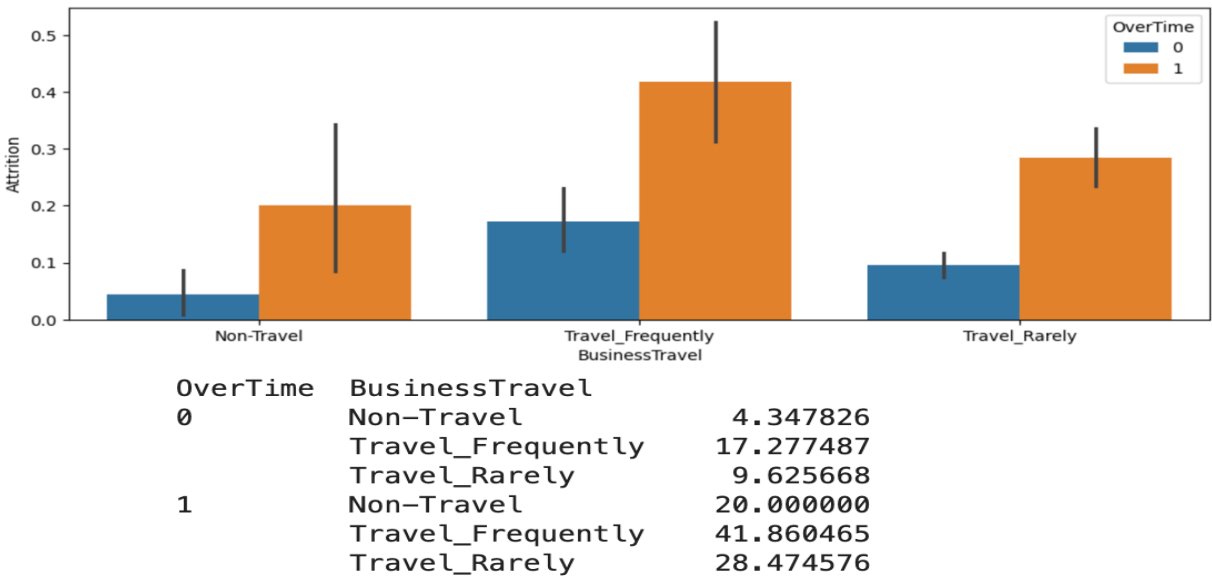
The bar chart shows that employees who travel frequently have the highest attrition rate at 24.9% — nearly 3 times higher than employees with no travel history (8.0%). Employees who travel rarely fall in between with an attrition rate of 14.95%.



Employees who travel frequently spend most of their time away from home, leaving them with little time for family and personal life. This constant travel leads to physical exhaustion and destroys their work-life balance, ultimately pushing them towards resignation.

Over time and business travel combinedly effect on Attrition?

The grouped bar chart combines the effect of overtime and business travel on attrition. Employees who both work overtime and travel frequently have the highest attrition rate at 41.86% — nearly 10 times higher than employees who neither work overtime nor travel (4.35%). This highlights that the combination of these two factors creates an extreme work burden on employees.



The combination of overtime and frequent travel creates extreme work pressure on employees. This demanding lifestyle leaves them with little to no time for personal life and family, leading to burnout and dissatisfaction. As a result the employees actively seek alternative opportunity that offer them better work life balance.

Key Finding

- Overall attrition rate is 16.1% above the industry average.
- Sales department has the highest attrition rate at 20.6%
- Employees working overtime are 3x more likely to resign (30.5%)
- Frequent travelers have nearly 3x higher attrition than non-travelers (24.9% vs 8%)
- Overtime + frequent traveler combined leads to 41.86% attrition 10x higher than those with neither.
- Single employees resign most (25.53%) due to fewer responsibilities.
- Younger employees (18-25) have the highest attrition at 35%
- Low job satisfaction doubles the attrition rate (22.83% vs 11.32%)
- Resigned employees earned 30% less than those who stayed (\$4787 vs \$6833)
- Junior level employees have the highest attrition 26.34%

Recommendations

- **Overtime Policy:** Make overtime voluntary and provide additional compensation for overtime hours to reduce burnout.
- **Business Travel:** Reduce unnecessary travel by utilizing virtual meetings where possible.
- **Junior Compensation:** Increase salary for entry-level employees to improve retention.
- **Young Employees:** Provide clear career growth paths, mentorship programs, and performance-based bonuses to retain young talent.
- **Sales Department:** Revise the compensation structure by introducing stable base salaries alongside commission.
- **Job Satisfaction:** Conduct regular employee feedback sessions to identify and address dissatisfaction early.

Conclusion

This analysis examined IBM HR data to identify the key drivers of employee attrition. The findings reveal that overtime, frequent business travel, low compensation, and job dissatisfaction are the primary factors contributing to the overall attrition rate of 16.1%. By providing additional compensation for overtime, reducing unnecessary business travel, offering clear career growth paths, and conducting regular employee feedback sessions, the organization can effectively address these issues and build a more stable and satisfied workforce.